

Crews from Evidence

Privacy-preserving, decentralised measurement of language-model teams,
and a cost-aware learner that picks them

Santosh Kumar Radha

AgentField AI

September 2026

Abstract

An agent harness runs a task through a *crew*: an assignment of one language model to each of n roles (in our instance a worker that does the work, a checker that reads it and a planner that cuts it). The crew that pays best is neither the cheapest nor the strongest on paper; it is the one that minimises the expected cost of one clean delivery once failed runs, wall time and shipped defects are priced. We give that objective in closed form, show how a crew is computed from a published catalog of prices and benchmark indexes alone (a pareto front in bill and quality, with a scale-free knee), and then how measurements from real runs replace the catalog's proxy. Runs are scored role by role by a model outside the crew. Each score becomes a row that carries no text, no file name and no identity beyond a per-install nonce, so nothing an install sends can be read back into what it was working on; rows reduce to sufficient statistics that form a join-semilattice, so any number of installs and a relay can merge them in any order and reach the same state, the relay is a convenience rather than an authority, and the published index is signed so that every reader can check what it is blending in. Participation is a setting: an install can contribute, read only, or stay out, and still learns from its own runs. Judge severity is removed by a two-way fixed-effects fit on those statistics, drift by a local-level model whose forgetting rate is estimated rather than set, and heterogeneity between installs by a hierarchical prior whose variance is estimated by moments; every constant except the user's price of a defect is an empirical-Bayes estimate. The crew is chosen by Thompson sampling with semi-bandit feedback: because the objective is linear in role qualities and every role is scored separately, a run on one crew informs every crew that shares a model with it, and the sample complexity scales with the number of models rather than the number of crews. On our own 120 scored runs the learner beats a fixed table by a third of the remaining regret within a few hundred runs, and role-level feedback cuts the regret of a crew-level learner by a further factor of three at four hundred runs.

Keywords: model selection, pareto front, empirical Bayes, Thompson sampling, combinatorial semi-bandits, CRDT, judge calibration, privacy, decentralised measurement.

1 Introduction

A harness that drives language models through software tasks has to decide, before every task, which model sits in which role. The decision is made under three kinds of information that arrive at different times and in different units: published prices and benchmark indexes, available for every model on the day it appears; the harness's own past runs, few but on the right distribution; and the runs of everyone else using the same harness, many but on other people's tasks and scored by other people's judges. This note is about turning the three into one decision.

Our contributions are the following.

1. A closed-form objective for a crew, the expected cost of one clean delivery under reruns and a priced defect (Section 4), and the observation that on a catalog alone the objective cannot be evaluated, which is why the catalog picker is a pareto front with a scale-free knee (Section 3).
2. A decentralised measurement of role quality (Section 5): text-free rows, sufficient statistics that form a state-based CRDT (Proposition 2), judge fixed effects identified from the connectivity of the judge-cell graph (Proposition 3), robust aggregation across installs, and drift handled by a local-level model.

3. An estimator with no tuned constant except the user’s price of a defect (Section 6): the catalog’s prior weight, the between-install variance, the noise scale and the forgetting rate are all estimated from the pool by moments.
4. A picker (Section 7) that is Thompson sampling in a combinatorial semi-bandit: the crew objective is linear in role qualities, roles are scored separately, so one run updates every crew sharing a model with it.
5. An empirical check on our own measured runs (Section 8), reproducible from the script beside this document.

2 Setting

Definition 1 (Roles, crews). Let $\mathcal{R}$ be a finite set of n roles and $\mathcal{M}$ a finite set of models. A *crew* is a map $\kappa : \mathcal{R} \rightarrow \mathcal{M}$. Each role r carries a *shape*: a token share $v_r \geq 0$ with $\sum_r v_r = 1$, index weights $w_r \in \Delta^2$, a cache share $\gamma_r \in [0, 1]$, an input-to-output ratio $\rho_r > 0$, and a predicate $\text{fit}_r(m)$ saying whether model m can hold the role (tools, images, context). A symmetric relation $D \subseteq \mathcal{R} \times \mathcal{R}$ names the pairs of roles that may not be held by models of the same vendor.

Our instance has $\mathcal{R} = \{\text{W}, \text{H}, \text{M}\}$ (worker, high, mastermind), token shares $(0.90, 0.08, 0.02)$, weights $(0.2, 0.3, 0.5)$, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $(0.6, 0, 0.4)$ on the indexes (intelligence, coding, agentic), cache shares $(0.75, 0.75, 0)$, ratios $(100, 100, 10)$ and $D = \{(\text{W}, \text{H})\}$: the two roles that read the same work never share one shop. On the chat door the planner role is absent and $n = 2$. Nothing below depends on $n = 3$.

A *task* is one unit of work; a *run* is one attempt at it by a crew. A run ends *delivered* or not, and a delivered task is later found *clean* (no major defect) or not. Money is in dollars and time is priced at λ dollars per minute.

The *catalog* lists, per model, a prompt price π_m , a cache-read price π_m^c , a completion price κ_m per million tokens, and benchmark indexes $I_m \in \mathbb{R}^3$. The *pool* is the set of measurements described in Section 5; the *own sheet* is one install’s part of it.

3 The crew computed from the catalog

Role cost and role quality. Per million task tokens on role r , model m costs

$$c_r(m) = (1 - \gamma_r) \pi_m + \gamma_r \pi_m^c + \kappa_m / \rho_r, \quad (1)$$

and its catalog quality is the weighted, pool-normalised index

$$\hat{q}_r(m) = 100 \sum_{i=1}^3 w_{r,i} \frac{I_{m,i}}{\max_{m' \in \mathcal{M}} I_{m',i}}. \quad (2)$$

A missing index is imputed from the median ratio, over models publishing both, of the missing index to a present one; an imputed quality is flagged and loses ties.

Shortlists. Role r ’s shortlist is

$$S_r = \{m : \text{fit}_r(m) \wedge \hat{q}_r(m) \geq \phi \max_{m'} \hat{q}_r(m')\}, \quad \phi = 0.8, \quad (3)$$

the floor being the statement that a model which cannot do the job is not cheap at any price. The crew set is $\mathcal{K} = \{\kappa \in \prod_r S_r : \text{vendor}(\kappa(r)) \neq \text{vendor}(\kappa(r')) \forall (r, r') \in D\}$.

Bill, quality, front.

$$B(\kappa) = \sum_r v_r c_r(\kappa(r)), \quad Q(\kappa) = \sum_r v_r q_r(\kappa(r)), \quad v_r \geq 0, \sum_r v_r = 1, \quad (4)$$

with v uniform by default. The *front* $\mathcal{F} \subseteq \mathcal{K}$ is the set of crews not dominated in (B, Q) . Sorting $\mathcal{K}$ by $(B \uparrow, Q \downarrow, \text{imputed count} \uparrow, \text{id} \uparrow)$ and keeping a crew when its Q exceeds every kept crew’s yields $\mathcal{F}$ in $O(|\mathcal{K}| \log |\mathcal{K}|)$, and the tie order makes $\mathcal{F}$ a function of the set $\mathcal{K}$ rather than of any input order. In practice $|S_r|$ is a few dozen and $|\mathcal{K}|$ a few thousand.

Why a front and not a score. The objective of Section 4 needs a probability of clean delivery. The catalog has none, so any scalarisation of (B, Q) would fix an exchange rate between a price and a benchmark index that nobody has measured. Dominance is the only order that needs no such rate; every crew a user could sensibly want is on $\mathcal{F}$, which is small, and the remaining choice is one-dimensional along it.

Knee and knob. Index $\mathcal{F}$ by bill, $B_0 \leq \dots \leq B_1$. The three named picks are *frugal* = $\arg \min_{\mathcal{F}} B$, *max* = $\arg \max_{\mathcal{F}} Q$, and *balanced*, the crew farthest above the chord of the front in $(\ln B, Q)$:

$$\text{balanced} = \arg \max_{\kappa \in \mathcal{F}} \left[(Q(\kappa) - Q_0) - (Q_1 - Q_0) \frac{\ln B(\kappa) - \ln B_0}{\ln B_1 - \ln B_0} \right]. \quad (5)$$

Proposition 1 (Scale invariance). *The balanced crew is invariant under $B \mapsto aB$ for any $a > 0$ and under $Q \mapsto Q + b$.*

Proof. $\ln(aB) - \ln(aB_0) = \ln B - \ln B_0$ and $\ln(aB_1) - \ln(aB_0) = \ln B_1 - \ln B_0$; the bracket is unchanged. A shift of Q cancels in both differences. $\square$

The log scale is not cosmetic: catalog prices span two orders of magnitude, and the knee should not move when a currency changes or every vendor cuts prices by the same factor. A knob $k \in [0, 1]$ picks the best crew with $B \leq B_0(B_1/B_0)^k$, so equal steps in k are equal ratios in the bill.

4 The objective: the cost of one clean delivery

Let a run of crew κ cost c dollars and t minutes, deliver with probability r , and let a delivered task be clean with probability p ; a run that fails to deliver costs c_f and t_f .

Lemma 1 (Rerun until clean). *If every defect is caught and the task rerun, the expected cost to one clean delivery is*

$$E_0(\kappa) = \frac{(1-r)(c_f + \lambda t_f) + r(c + \lambda t)}{rp}. \quad (6)$$

Proof. Attempts are i.i.d. with success probability rp ; the count to the first success is geometric with mean $1/(rp)$, and by Wald's identity the expected total cost is the mean count times the expected cost of one attempt, which is the numerator. $\square$

Lemma 2 (Priced defect). *If a shipped defect costs H to find and mend and only non-deliveries are rerun,*

$$E_H(\kappa) = \frac{(1-r)(c_f + \lambda t_f)}{r} + (c + \lambda t) + (1-p)H. \quad (7)$$

Proof. The count of failed attempts before the first delivery is geometric on $\{0, 1, \dots\}$ with mean $(1-r)/r$; the delivering attempt costs $c + \lambda t$; the defect is paid once with probability $1-p$. $\square$

H is the one quantity that belongs to the user, and the budget word sets it: frugal $H = \$1$, balanced $H = \$10$, max $H = \$50$. Both forms are monotone in the same three quantities, the run's cost, its time and $P_c = rp$, and the picker works with

$$E(\kappa) = c(\kappa) + \lambda t(\kappa) + H(1 - P_c(\kappa)). \quad (8)$$

Figure 1 draws our measured crews with the iso-lines of (8). The knob k and the price H parametrise the same trade-off; k is what a catalog admits, H is what measurements admit.

Assumption 1 (Linearity in roles). $P_c(\kappa) = \sum_r \nu_r q_r(\kappa(r))/100$, where $q_r(m)$ is the pool's role quality: the run's outcome credited to the role, on a 0–100 scale.

The assumption is what makes (8) linear in the role qualities, which Section 7 uses. It is a working rule rather than a law: a product over roles would count one outcome n times, and a fitted interaction becomes available once judged scores separate the roles' contributions. Nothing in Sections 5–6 depends on it.

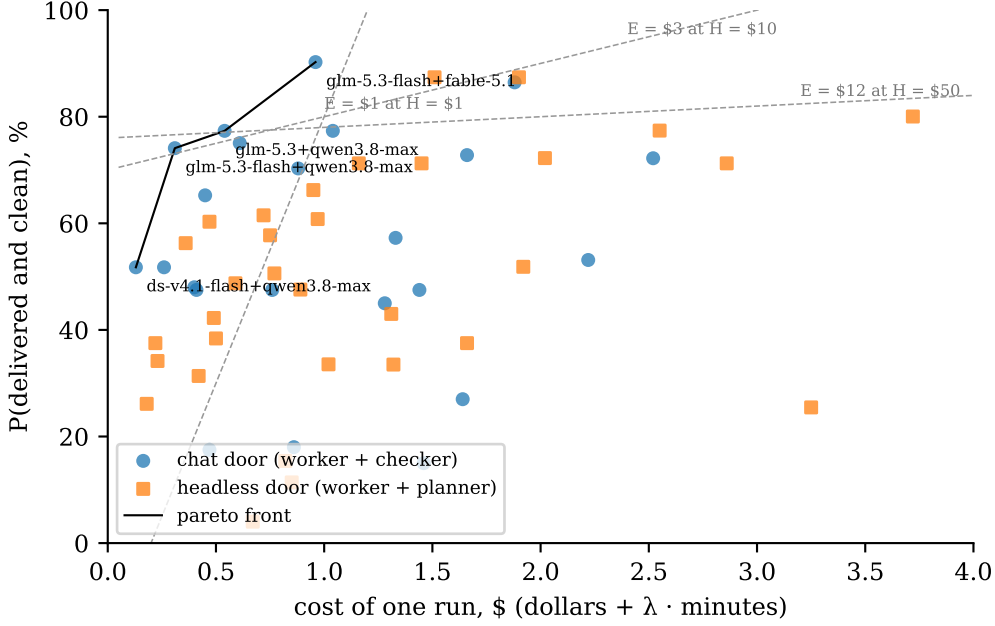


Figure 1: Our measured crews (56 combinations fitted from 120 scored runs): cost of one run against the probability that it delivers clean. Solid: the pareto front. Dashed: iso-cost lines of (8); a larger H tilts them so that quality buys more.

5 The Model Pool: a decentralised measure of role quality

5.1 Judged runs and rows

After a run finishes, one model outside the crew reads the record (brief, deliverable, the checker’s report and claim, the files touched) and answers one question per role with a score in $[0, 100]$ and one sentence. The judge is chosen deterministically: the cheapest catalog model with coding index at least 60 that can call tools, holds no role in the crew, and is not from the worker’s vendor; a role the judge’s own model holds is skipped. The floor exists because a judge below the worker’s class rubber-stamps. The judge’s calls are billed to a role of their own, so learning is never free by accident; at present prices it is cents per run.

One judged role of one run is one *row*

$$x = (\text{role}, \text{model}, \text{score}, \text{dollars}, \text{minutes}, \text{size} \in \{S, M, L\}, \text{door}, \text{judge}, \text{day}, \text{install}),$$

where *install* is a nonce drawn once per profile. No text of any kind is in a row. Rows wait in a local outbox, which can be inspected, until they are sent.

5.2 Sufficient statistics and order-free merging

Definition 2 (Cell, sheet). A *cell* is a tuple $\theta = (\text{role}, \text{model}, \text{judge}, \text{day}, \text{size}, \text{door})$. A *sheet* is a map from (install, cell) to a triple $T = (n, S, S^2) \in \mathbb{N} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$, the count, sum and sum of squares of the scores an install has produced in that cell.

Every quantity computed below is a function of the sheet. Sheets are merged by the join

$$(A \sqcup B)(u, \theta) = (\max(n_A, n_B), \max(S_A, S_B), \max(S_A^2, S_B^2)) \quad \text{taken componentwise on the triples } A(u, \theta), B(u, \theta), \quad (9)$$

with a missing entry read as $(0, 0, 0)$.

Proposition 2 (The sheet is a state-based CRDT). *Let each install u only ever publish its own entries (u, θ) and let every published triple be the running total of u ’s rows in θ . Then (i) $\sqcup$ is commutative, associative and idempotent; (ii) an install’s triples are monotone non-decreasing in time, so a later publication dominates an earlier one under $\sqcup$; (iii) any two nodes that have received the same set of publications, in any order and with any duplication, hold the same sheet.*

Proof. (i) is the componentwise max on a product of totally ordered sets. (ii) Scores lie in $[0, 100]$, so n , S and S^2 each grow by a non-negative amount per row. (iii) follows from (i) and (ii): the join of a set of publications is their least upper bound, which does not depend on order or multiplicity, and by (ii) the least upper bound of an install's publications is its latest. $\square$

The cell means and pooled variances are functions of $\sum_u T(u, \theta)$, so by Proposition 2 a relay and every install compute the same estimates from the same publications; there is no coordinator whose order of arrival matters, and a relay that is lost is rebuilt from any node that kept the publications. This is the sense in which the measure is decentralised: the relay is a convenience for bandwidth and signing, not a source of truth.

5.3 Judges are not the same ruler

Let y be a score of role r , model m , by judge j . We posit the two-way additive model

$$y = \mu_{rm} + \beta_j + \varepsilon, \quad \mathbb{E}\varepsilon = 0, \quad \text{Var } \varepsilon = \sigma^2, \quad (10)$$

with μ_{rm} the quantity of interest and β_j the judge's severity, constrained by $\sum_j n_{..j}\beta_j = 0$ (the judges' weighted mean is the ruler). Weighted least squares needs only the cell means $\bar{y}_{rmj} = S_{rmj}/n_{rmj}$ and counts. The normal equations are solved by alternating

$$\beta_j \leftarrow \frac{\sum_{r,m} n_{rmj}(\bar{y}_{rmj} - \mu_{rm})}{\sum_{r,m} n_{rmj}}, \quad \mu_{rm} \leftarrow \frac{\sum_j n_{rmj}(\bar{y}_{rmj} - \beta_j)}{\sum_j n_{rmj}}, \quad (11)$$

re-centring β after each sweep. Each half-step minimises a convex quadratic in its block, so the residual is non-increasing and the iteration converges to the least-squares solution; five sweeps suffice in practice.

Proposition 3 (Identifiability). *Let G be the bipartite graph with the judges on one side, the pairs (r, m) on the other, and an edge where $n_{rmj} > 0$. The parameters of (10) under the centring constraint are identified on a connected component of G ; across components only the within-component contrasts are.*

Proof. The design matrix of a two-way additive model has rank equal to (number of vertices of G) minus (number of connected components); one linear constraint per component fixes the remaining freedom. With the single centring constraint, exactly one component is fully identified. $\square$

The judge rule of Section 5.1 makes G connected quickly in practice: it sends most installs to the same few cheap judges. A model scored by a single judge so far has μ_{rm} equal to its own mean and the index says so.

5.4 Installs are not independent, and some are wrong

Rows from one install share a task mix, a codebase and a way of writing briefs. Under an intra-class correlation ϱ , n_u rows from install u carry the information of $n_u/(1 + (n_u - 1)\varrho)$ independent ones, which saturates at $1/\varrho$. The pool's cell estimate therefore treats installs, not rows, as the unit: it forms the install means $\bar{y}_{u\theta}$ and combines them with a location estimator that is robust to a minority of installs that are broken or adversarial. We use the Huber M -estimate with scale $s = 1.4826 \cdot \text{MAD}$ and the conventional tuning constant 1.345 (95% efficiency at the normal), which has breakdown point $1/2$ in location, and fall back to the precision-weighted mean when a cell has fewer than five installs. A cell backed by fewer than three installs is published but not read.

5.5 Drift

Models change under the same name. Let $\bar{y}_d$ be a cell's day- d mean over n_d rows and model the true level as a random walk,

$$\mu_d = \mu_{d-1} + \eta_d, \quad \eta_d \sim \mathcal{N}(0, \omega^2); \quad \bar{y}_d = \mu_d + \bar{\varepsilon}_d, \quad \bar{\varepsilon}_d \sim \mathcal{N}(0, \sigma^2/n_d). \quad (12)$$

The Kalman filter for this local-level model is, in steady state with constant $n_d = \bar{n}$, an exponentially weighted average whose forgetting factor $1 - \alpha$ solves $\alpha^2/(1 - \alpha) = \bar{n}\omega^2/\sigma^2$; the effective memory is about $1/\alpha$ days. The drift variance is estimated by moments from the day-to-day differences,

$$\hat{\omega}^2 = \frac{1}{2} \left[\widehat{\text{Var}}(\bar{y}_d - \bar{y}_{d-1}) - \sigma^2 \left(\frac{1}{n_d} + \frac{1}{n_{d-1}} \right) \right]_+, \quad (13)$$

pooled across cells. A cell with no evidence of drift ($\hat{\omega}^2 = 0$) keeps everything; a cell that drifts forgets at the rate the data show. The day is a cell dimension for exactly this reason.

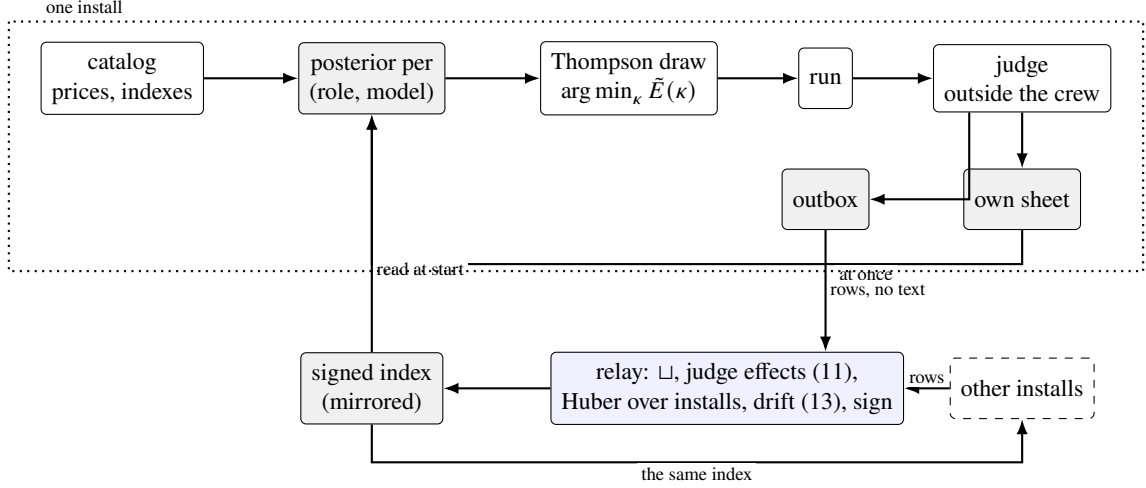


Figure 2: Where each step runs. Inside the dotted box is one install; the only thing that crosses it outward is a row without text. The relay merges publications with the join of Proposition 2 and publishes a signed index; any install can host a relay, and any node that kept the publications can rebuild one.

6 Estimation with no tuned constant

We now combine the three sources for one install u . Write $\hat{q}_r(m)$ for the catalog quality (2), $\bar{y}_{rm}^{\text{pool}}$ and n_{rm}^{pool} for the pool’s cell after Sections 5.3–5.5, and $\bar{y}_{rm}^u$, n_{rm}^u for the own sheet.

Assumption 2 (Hierarchy).

$$\mu_{rm} \sim \mathcal{N}(\hat{q}_r(m), \tau_0^2), \quad \mu_{rm}^u \mid \mu_{rm} \sim \mathcal{N}(\mu_{rm}, \tau_1^2), \quad y \mid \mu_{rm}^u \sim \mathcal{N}(\mu_{rm}^u, \sigma^2). \quad (14)$$

The first line says the catalog is an unbiased but noisy predictor of the population quality; the second that install u ’s own quality for the model scatters around the population’s; the third is judge noise after Section 5.3. The install cares about μ_{rm}^u .

Proposition 4 (Posterior for the install). *Under (14), with $k = \sigma^2/\tau_0^2$ and $\ell = \sigma^2/\tau_1^2$, the posterior of μ_{rm}^u is normal with*

$$\mathbb{E}[\mu_{rm}^u \mid \cdot] = \frac{k^* \hat{q}_r(m) + n^* \bar{y}_{rm}^{\text{pool}} + n^u \bar{y}_{rm}^u}{k^* + n^* + n^u}, \quad \text{Var}[\mu_{rm}^u \mid \cdot] = \frac{\sigma^2}{k^* + n^* + n^u}, \quad (15)$$

where the pool and the catalog enter with the capped effective counts

$$n^* = \frac{n^{\text{pool}} \ell}{n^{\text{pool}} + \ell} \leq \ell, \quad k^* = \frac{k \ell}{k + \ell} \leq \ell. \quad (16)$$

Proof. Integrate out μ_{rm} . The catalog and the pool inform μ_{rm}^u only through μ_{rm} , whose posterior given them is normal with precision $(k + n^{\text{pool}})/\sigma^2$; adding the between-install variance τ_1^2 gives, for μ_{rm}^u , a precision $1/(\sigma^2/(k + n^{\text{pool}}) + \tau_1^2)$. Writing that as an effective count over σ^2 gives $(k + n^{\text{pool}})\ell/(k + n^{\text{pool}} + \ell)$, which splits into the two terms of (16) in proportion to k and n^{pool} ; the own rows add precision n^u/σ^2 directly. Precision-weighting the three means gives (15). $\square$

Corollary 1. *With $\tau_1 = 0$ ($\ell = \infty$) the posterior mean is $(1 - w)\hat{q}_r(m) + w\bar{y}$ with $w = n/(n + k)$, $n = n^{\text{pool}} + n^u$: the plain shrinkage blend, which is what the picker applies today with $k = 30$.*

The cap (16) is the point of the hierarchy. However many rows the pool holds, its evidence about *this* install’s quality is worth at most $\ell = \sigma^2/\tau_1^2$ own rows, because other installs’ tasks are not this install’s tasks; and the own rows are never capped. This replaces an ad hoc “own rows count triple” by a quantity with a meaning and an estimator.

Estimating the constants. Every constant in (15) is estimated from the sheet by moments; none is set by hand.

$$\hat{\sigma}^2 = \frac{\sum_{u,\theta} (S_{u\theta}^2 - S_{u\theta}^2/n_{u\theta})}{\sum_{u,\theta} (n_{u\theta} - 1)} \quad \text{pooled within-cell variance,} \quad (17)$$

$$\hat{\tau}_1^2 = \left[\widehat{\text{Var}}_u(\bar{y}_{u\theta}) - \hat{\sigma}^2 \overline{(1/n_{u\theta})} \right]_+ \quad \text{between-install variance, pooled over cells,} \quad (18)$$

$$\hat{\tau}_0^2 = \left[\widehat{\text{Var}}_{(r,m)}(\bar{y}_{rm}^{\text{pool}} - \hat{q}_r(m)) - \hat{\sigma}^2/n_{rm}^{\text{pool}} \right]_+ \quad \text{how far the catalog misses,} \quad (19)$$

each a within-minus-noise moment estimator of the kind used by Efron and Morris [6]. Then $k = \hat{\sigma}^2/\hat{\tau}_0^2$ and $\ell = \hat{\sigma}^2/\hat{\tau}_1^2$, and $\hat{\omega}^2$ of (13) sets the forgetting. These are published in the index beside the cells so that every install blends with the same constants, and re-estimated at every relay cycle. The relay withholds an estimate until its denominator has at least ten degrees of freedom and publishes the day-one values of Table 1 until then.

6.1 Covariates, and why no row is ever wasted

A row carries more than (role, model, score): a door, a size bucket, a judge, a day, an install, and any field added later, such as the harness version or a task family. Call the vector of such fields the row’s *profile* $c \in C = \prod_f L_f$, a product of factor levels. The hierarchy (14) extends factor by factor:

$$\mu_{rm,c} = \mu_{rm} + \sum_f \delta_{rm,c,f}^f, \quad \delta_{rm,\cdot}^f \sim \mathcal{N}(0, \tau_f^2) \text{ i.i.d. over levels,} \quad (20)$$

so that a (role, model) has one population quality and one partially pooled offset per level of each factor. The install factor is the case $f = \text{install}$ of (20), and Proposition 4 is its two-level instance; the same derivation gives, for a query profile c , a posterior mean that is a precision-weighted combination of the cell means at every level that shares a coordinate with c , each capped by its own $\ell_f = \sigma^2/\tau_f^2$, with $\hat{\tau}_f^2$ estimated by the between-level moment (18) applied to factor f . Two properties follow.

Old rows keep their value when the harness changes. If the door algorithm is rewritten, rows from the old door are a level of the door factor; they still inform μ_{rm} with the full weight the estimated τ_{door}^2 allows, and the new door starts at μ_{rm} plus nothing and specialises as its own rows arrive. If the two doors turn out to differ little, $\hat{\tau}_{\text{door}}^2 \approx 0$, $\ell_{\text{door}} \rightarrow \infty$, and the factor pools completely; if they differ a lot, the cap keeps the old door from speaking for the new one. The data decide, per factor, how much they transfer.

The schema is open. A row with a field absent (older rows, before the field existed) is at a level of its own, *unknown*, which the estimator treats like any other level. Adding a field therefore invalidates nothing already collected, and the sheet’s join in Proposition 2 is unaffected because a cell key is only a tuple. What can be extracted from the rows is the set of contrasts the factors expose; a factor that is never recorded cannot be recovered later, which is why the row carries every cheap field now and the text of none.

Regressors that are not factors, a task’s token count for instance, enter the same way as a linear term with a partially pooled slope; the moment estimators are the same with a design column in place of an indicator.

Interactions: the crew as a group. Assumption 1 scores roles one at a time. Whether a particular worker and a particular checker do better *together* than their role qualities predict is a question about the pair, and the same hierarchy answers it: a pair $(m_r, m_{r'})$ is a level of a factor over $\mathcal{M} \times \mathcal{M}$, with its own partially pooled offset $\delta_{(m_r, m_{r'})}^{rr'} \sim \mathcal{N}(0, \tau_{rr'}^2)$. A pair that has never been run sits at zero offset and the additive prediction stands; a pair with rows moves by exactly the amount its rows justify against $\hat{\tau}_{rr'}^2$; if pairs in general add nothing, $\hat{\tau}_{rr'}^2 \rightarrow 0$ and the term disappears. This is the whole difference between learning crews as groups and learning roles: the group term is there, but it is charged for. A crew-level learner with no structure has to visit each of $|\mathcal{K}|$ crews to say anything about it; the additive model needs $\sum_r |S_r|$ observations to say something about every crew, and the pair term then spends further observations only where a pair is actually run. Some group effects are constraints rather than parameters: the vendor rule D is one, and it is applied to $\mathcal{K}$ before any learning happens.

7 Deciding: Thompson sampling with semi-bandit feedback

Given a posterior $\mathcal{N}(\mu'_{rm}, \sigma'^2_{rm})$ per role and model from (15), the pick could minimise (8) at the posterior means. That never tries a crew the prior underrates, and the catalog prior is wrong by a lot for some models (Figure 6). Thompson sampling [1, 2] draws one plausible world from the posterior and acts as if it were true: a crew is chosen

constant	meaning	estimator	day-one value
σ	judge noise	(17)	15
$k = \sigma^2 / \tau_0^2$	weight of the catalog	(19)	30
$\ell_f = \sigma^2 / \tau_f^2$	cap on evidence transferred across a factor (install, door, size, ...)	(18) per factor	30
ω^2	drift per day	(13)	0 (no forgetting)
β_j	judge severity	(11)	0
v_r	role token shares	measured from run token accounting	0.90/0.08/0.02
H	price of a shipped defect	the user's, by budget word	\$1 / \$10 / \$50
λ	price of a minute	the user's	\$0.01
ϕ , judge floor	structural floors	fixed by design	0.8, 60

Table 1: Every constant with its estimator. Two are the user's by design; two are structural floors. The rest are empirical-Bayes estimates that the relay publishes with the index.

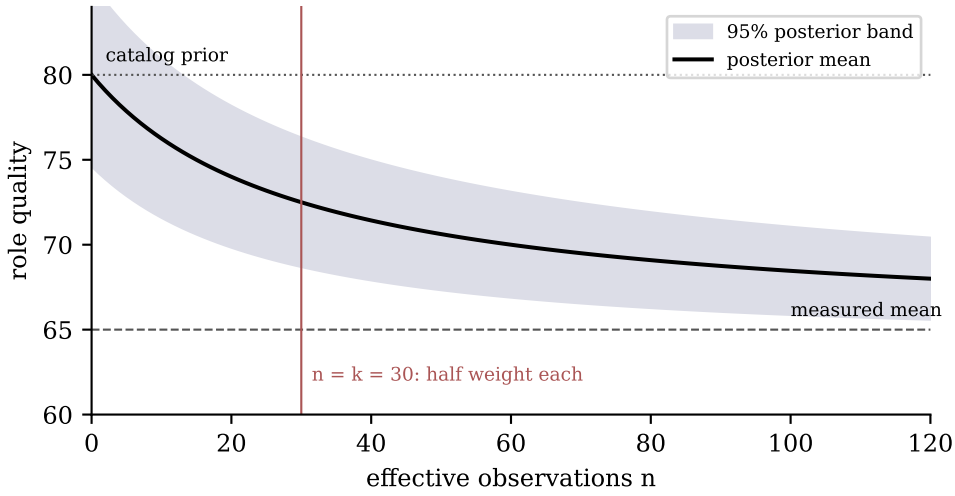


Figure 3: Shrinkage of one role quality from the catalog prior (80) toward the measured mean (65), $k = 30$, $\sigma = 15$. The band is what the Thompson draw of Section 7 samples from.

with the probability that it is the best, exploration is proportional to what is unknown, and it stops on its own as the posteriors tighten. No exploration rate exists to be set.

Proposition 5 (Role draws are crew draws). *Under Assumption 1, drawing $\tilde{q}_{rm}$ independently per (role, model) and minimising $\tilde{E}$ over $\mathcal{K}$ is exactly Thompson sampling over the crews' posterior costs.*

Proof. $\tilde{E}(\kappa)$ is an affine function of the vector $(\tilde{q}_{rm})$, so its joint law over $\mathcal{K}$ is the push-forward of the posterior on role qualities; a sample of the latter is a sample of the former. $\square$

The structure that matters is the feedback. A run of crew κ is judged role by role, so it yields one observation for each of its n (role, model) pairs, and the posterior of every crew that shares a pair with κ moves. This is a combinatorial bandit with semi-bandit feedback [4, 5]: the unknowns are the $\sum_r |S_r|$ role qualities, not the $|\mathcal{K}|$ crew values, and the regret of Thompson sampling scales with the former [5]. Section 8 measures the difference on our runs. For Gaussian posteriors the Bayesian regret of Thompson sampling is $O(\sqrt{dT \log T})$ in the dimension d of the unknowns over T runs [3].

The learned front. Under *learn* the front of Section 3 is recomputed on the posterior: the bill $B(\kappa)$ is read at the run's expected tokens, and the quality is the posterior mean of $P_c(\kappa)$, so the front is the pareto set in (expected dollars per task, expected probability of a clean delivery) with an interval on each point from (15). The budget word's H is a line of slope $-1/H$ in that plane and the pick is where the line first touches the front; Algorithm 1 is the same pick on one draw of the front rather than on its mean, which is what makes it explore. The pool command

Algorithm 1 One pick under *learn*

Require: shortlists S_r on the posterior means, budget word $\Rightarrow H$, task size bucket $\Rightarrow$ expected tokens T and minutes $t(\kappa)$

- 1: **for** each role r and $m \in S_r$ **do**
- 2: $\tilde{q}_{rm} \sim \mathcal{N}(\mu'_{rm}, \sigma'^2_{rm})$ ▷ one draw per run; never redrawn during the run
- 3: **end for**
- 4: **for** each $\kappa \in \mathcal{K}$ **do**
- 5: $\tilde{E}(\kappa) \leftarrow T B(\kappa) + \lambda t(\kappa) + H (1 - \sum_r v_r \tilde{q}_{r\kappa(r)}/100)$
- 6: **end for**
- 7: **return** $\arg \min_{\kappa} \tilde{E}(\kappa)$ ▷ ties: fewer imputed indexes, then id order

prints the mean front with its intervals, and the catalog front beside it, so that a user can see where measurement moved a pick and by how much.

When is it settled. The same posterior answers whether learning is done: the probability π^* that the posterior-mean crew is the best, estimated by repeating the draw of Algorithm 1 a few hundred times. The pool command reports π^* ; a value near one means further exploration is nearly free, a value near $1/|\mathcal{F}|$ means the ranking is still open. Costs enter as observed dollars per run and are not sampled; their noise is small next to the quality's.

Remark 1 (A new model). A model with no rows has $n = 0$ and sits at its catalog prior with the prior variance σ^2/k^* . If that puts it near the front it is drawn into crews soon, judged, and confirmed or pushed down by its first rows. Nothing is done by hand when the catalog changes.

8 Experiments on our measured runs

Ground truth is the table of Figure 1: 56 crews with P_c , dollars and minutes fitted from 120 scored runs of our own. Each replicate draws a catalog prior per arm as the truth plus $\mathcal{N}(0, 0.15^2)$ noise, roughly the disagreement between indexes and measurements in Figure 6, holds it with $k = 30$, and runs N tasks; a strategy picks, observes the outcome and pays (8) with $H = \$10$. Regret is the excess over the oracle crew, averaged over 200 replicates.

Crew-level learners. Table 2 and Figure 4 compare, on the twelve chat-door crews, a fixed table row, the catalog prior alone, ε -greedy and Thompson, all with one arm per crew. After 300 runs both learners are about \$0.30 per task under the fixed table, a third of the table's regret, and still falling. Thompson pays a small exploration tax in the first twenty runs; $k = 30$ is what keeps it small. ε -greedy matches it here only because $\varepsilon = 0.1$ happens to suit a dozen arms; it neither grows when a new model appears nor shrinks when the ranking is settled.

N	10	20	40	70	100	150	200	300	400
table	0.91	0.93	0.96	0.94	0.94	0.93	0.94	0.96	0.97
catalog	0.82	0.86	0.88	0.87	0.85	0.83	0.83	0.84	0.86
greedy	0.72	0.79	0.79	0.78	0.76	0.72	0.69	0.64	0.62
thompson	0.98	0.97	0.90	0.86	0.81	0.75	0.70	0.65	0.59

Table 2: Regret against the oracle, dollars per task after N runs, one arm per crew.

Role-level feedback. Figure 5 and Table 3 isolate the structure of Section 7. The ground truth is three-role: a role quality per (role, model) read off the measured rows (a worker's delivered rate, a checker's and a planner's clean rate), combined by the linear rule of Assumption 1 over every crew the vendor rule admits, with dollars additive over roles. Three learners run Thompson sampling with the same prior noise and the same H and differ only in what they see after a run: one arm per crew fed the run's clean-or-defect bit; one arm per crew fed the mean of the judge's role scores; one posterior per (role, model) fed each role's own score, which is what the judge of Section 5.1 provides. The last is the semi-bandit. Richer feedback per run helps (the second learner over the first), but the structure helps more: at four hundred runs the role-level learner's regret is about a third of the crew-level learner's on the same scores, because every run updates the crews that share a model with the one that ran, and there are sixteen unknowns behind 138 crews.

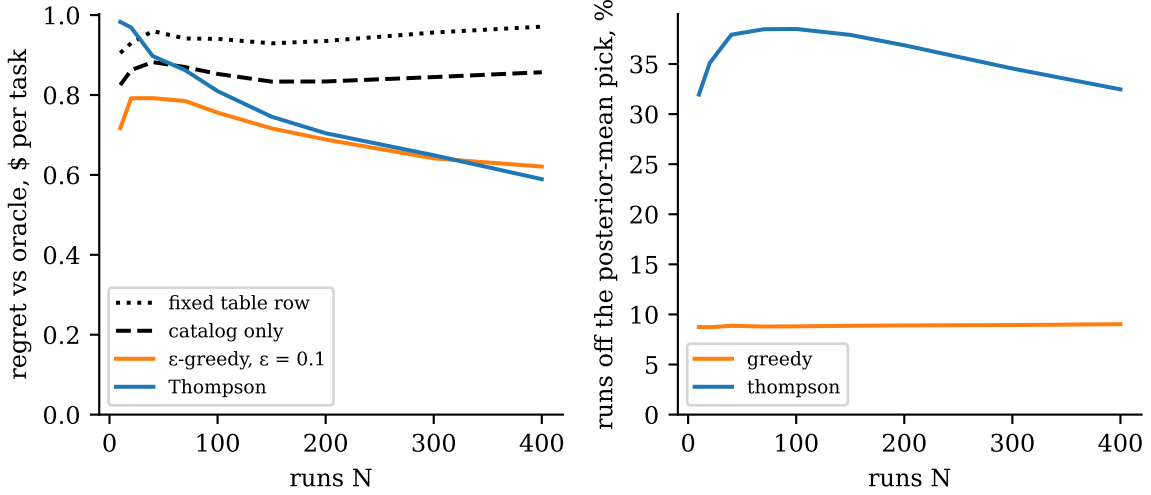


Figure 4: Left: regret per task. Right: share of runs on a crew other than the current posterior-mean best. Thompson explores heavily while a dozen crews sit within cents of each other and less as the posteriors separate; ϵ -greedy explores at its fixed rate.

N	10	20	40	70	100	150	200	300	400
crew, clean bit	0.47	0.50	0.50	0.51	0.53	0.54	0.55	0.55	0.54
crew, mean score	0.29	0.43	0.39	0.38	0.36	0.35	0.34	0.34	0.31
role, semi-bandit	0.36	0.35	0.31	0.27	0.26	0.22	0.20	0.17	0.15

Table 3: Regret against the oracle, dollars per task after N runs, on the three-role truth; same prior noise and H for all three learners.

The truth in these experiments is our own fitted table from one team’s tasks, so the size of the gain is the shape of the effect and not a promise for every install. The script `figures.py` beside this document reproduces every figure and table from the two CSV files it reads.

9 What the user controls

setting	values	meaning
crew	frugal balanced max custom	how much to spend: H and the knob k
picked from	table catalog learn	the rows we measured; recomputed from today’s catalog; catalog plus pool plus own sheet
model_pool	on read off	read the index and send rows (default); read only; neither
explore	on off	Thompson draw (default) or the posterior mean; <i>learn</i> only

With the pool set to *read*, the own sheet still accrues and enters (15); with it *off*, only the catalog prior remains and the pick is the catalog’s. The index is a JSON document signed with an Ed25519 key whose public half is in the binary; `codeaf pool verify` checks it, `pool show` prints it, `pool status` lists the rows waiting in the outbox. The binary carries a seed index of our own measured runs (Figure 6) so that *learn* has numbers on day one. The relay’s code is in this repository; its addresses are settings, so an install can read from and send to a relay of its own, or to none.

10 Status

On the branch this document ships with: the catalog crew of Section 3; the shrinkage blend of the Corollary with $k = 30$; the index document, its reader and signed pull; the outbox and the sheet; the judge and its deterministic choice; the settings; the seed index. Specified here and next to land: the judge fixed effects, Huber aggregation and drift estimate in the relay; the hierarchical cap (16) and the moment estimators (17)–(19); the Thompson draw with

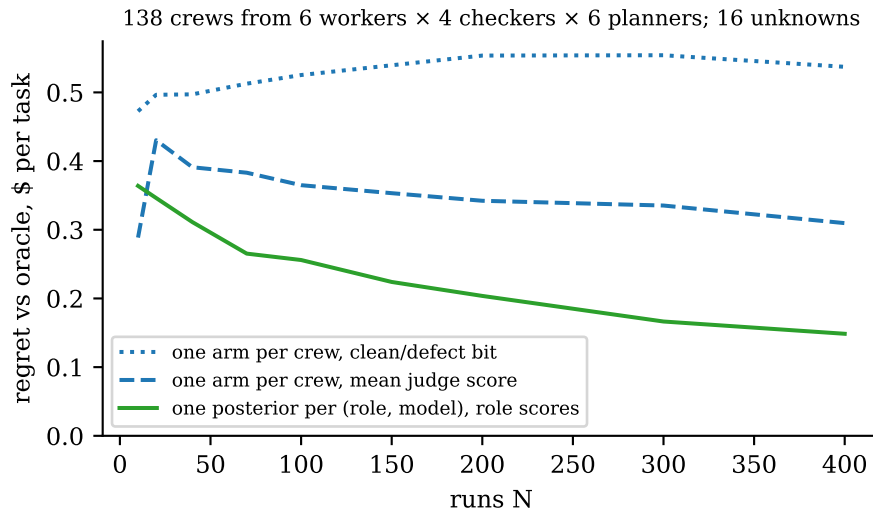


Figure 5: Thompson sampling under three kinds of feedback on the same three-role truth. Dotted and dashed: one arm per crew, fed the clean bit and the mean role score respectively. Solid: one posterior per (role, model) fed the role scores, the combinatorial semi-bandit of Section 7.

role-level updates and the settled probability π^* ; the relay itself. Until the relay exists the pool is the seed index plus each install’s own sheet.

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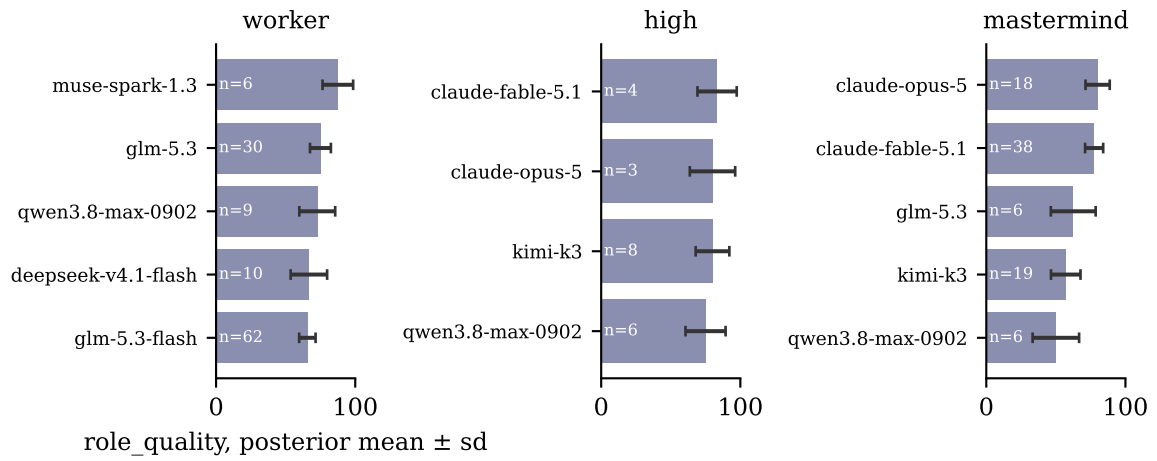


Figure 6: The seed index shipped in the binary: 14 cells from 120 scored runs, $\text{role_quality} = 100 \cdot P(\text{no major defect})$ under a $\text{Beta}(1,1)$ posterior. The spread between models a catalog rates alike is why the pool exists.